

BlueYonder VMware Environment

Discovery Notes from Red Hat OVE Demo Call

Source call: April 21, 2026 | Attendees: Scott Jobe, Todd Bowen, Kelly Nowell, Dan Wilson (BlueYonder); John Kinney, Michael Quick, Bill Venella, Joe Moreno (Red Hat)

The following details about BlueYonder's current VMware environment were captured from the Red Hat OpenShift Virtualization Engine (OVE) demo call. These are extracted directly from customer comments during the session and are intended as a reference for follow-up planning.

Workload Profile

- Currently running VMs on VMware; evaluating OpenShift Virtualization as a potential replacement.
- Initial scope is lift-and-shift of VMs only — not containers, though the team wants to understand future container pivot options.
- Previously evaluated full OpenShift, but pricing came close to what VMware was charging. They are now revisiting via the cheaper OVE-only path.

Legacy and Sensitive Applications

- Legacy apps *"written in the 70s"* that are extremely latency-sensitive and will *"freak out if they lose a ping."*
- Specifically called out: **Oracle databases** and **SQL databases** as workloads with strict performance and latency requirements.
- These sensitive workloads are *not* hosted in dedicated clusters. They coexist with other VMs but are protected via affinity / anti-affinity rules combined with performance profiles, so other VMs are evicted from the host first.

DRS and Scheduling Configuration

- Actively use DRS with custom-tuned balancing thresholds (described informally as "lazy Larry vs aggressive Andy").
- Heavy use of affinity and anti-affinity rules to keep sensitive workloads pinned and protect them from migration impact.

Storage Configuration

- Use **PV SCSI** controllers for high-IO secondary disks (databases).
- Use built-in **LSI** drivers only on certain boot disks where the vendor requires it.
- Significant in-OS tuning has been done to work around SCSI controller IO splitting limitations — disks are deliberately distributed across multiple SCSI controllers, since a controller with fewer disks runs faster than one with many.
- Substantial time has been spent over the years tuning storage layout in collaboration with the DBAs.
- Note: a 30,000 transactions-per-second figure was discussed on the call, but that came from a Red Hat reference (NASA JPL), not from BlueYonder. BlueYonder did not state their own TPS number.

Networking Configuration

- All VLANs in the environment are automatically exposed to every ESXi host.
- The network team creates VLANs centrally and they propagate downstream automatically — admins do not manually configure VLANs per switch.
- Adding a new VM network is essentially: cluster switch → new network → VLAN tag → save.
- BlueYonder is **not using NSX** (confirmed from a prior call referenced by John Kinney).
- No micro-segmentation requirements were raised on this call.

Operational Practices

- Day-to-day administration and troubleshooting are primarily GUI-driven.
- Scripting is used for large-scale operations — primarily **PowerShell**, with some Bash. Example use case: changing settings across roughly 200 VMware datastores.
- Ansible is in the environment but barely used — limited to some VM-build automation by the build team, and not used directly against VMware.
- Automation in on-prem data centers is limited overall. Scott noted this means a platform pivot would not be a massive lift from an automation rewrite perspective.
- Multi-cluster environment — the team specifically asked about cross-cluster live migration, noting most competitors require a power-off for that scenario.

Organizational Context

- **Scott Jobe** — lead infrastructure / VMware decision-maker on the call.
- **Todd Bowen** — handles deeper VM, storage, and DBA-adjacent work.
- **Kelly Nowell** and **Dan Wilson** — also on the BlueYonder team.
- BlueYonder governance is expected to block sending staff to Red Hat Summit in Atlanta unless the product has been purchased.

Identified Follow-up Items

- Detailed walkthrough of OpenShift Virtualization licensing vs. full OpenShift suite licensing.
- Confirm what is included with OVE vs. what would be an add-on (e.g., ACM for cross-cluster live migration, advanced networking).
- Review storage / IO tuning guidance specific to OpenShift Virtualization (Red Hat to share VirtIO / IO threads blog post).
- Deeper-dive session on networking — node network configuration policies, OVS bridges, and network attachment definitions.